The Catalogue Is the Instrument:
How the choice of sign inventory moves the statistics of an undeciphered script

Ilpo Väätäinen*

Version 3, 23 August 2026

Version history. *v2 (23 August 2026)*: §7 gained four collections of native Rapa Nui prose supplied by E. Korovina, one of which dissents for reasons of transcription rather than language. *v3 (23 August 2026)*: Appendix A is rewritten. Versions 1 and 2 reported rongorongo's repetition rates against a shuffle of the *pooled* corpus and concluded that a plain syllabic reading was ruled out. Against a null matched to the way the reference languages were measured—one shuffle per text—*AA* falls from 1.6–2.4 to 1.12, *AAA* from 5–15 to 1.68, and *ABAB* from 30–100 to 20.7, and Linear B, an actual syllabary, exceeds rongorongo on *AAA*. **The elimination argument does not survive and has been withdrawn.** What survives is stated in its place. §2 also notes two CEIPP codes that are not signs and were counted as signs in two of the three readings; correcting that moves the headline ratio from 4.14 to 4.28.

Abstract

Statistical arguments about undeciphered scripts—whether a sign system "behaves like language"—are computed on a transliteration, and a transliteration presupposes a catalogue of signs. For rongorongo the published catalogues disagree by an order of magnitude (about 50 to about 600 basic signs). Pozdniakov and Pozdniakov [2007] said in 2007 that statistics "will differ markedly depending upon the inventory of glyphs chosen"; the point has not, to our knowledge, been quantified. We quantify it on the CEIPP corpus (25 objects, 14 812 coded glyphs) using the normalised conditional entropy of Rao et al. [2010], on whose $[0, 1]$ scale the language-versus-non-language comparisons are drawn. Three published readings of the *same* inscriptions move the statistic by 0.318 of that scale; the corpus's distance from its own unigram-matched shuffle—the structure the statistic is meant to detect—is 0.077. The choice of catalogue therefore moves the instrument about four times further than the signal (3.7× after Miller–Madow correction). A sweep over catalogue sizes from 25 to 633 under four merge rules shows the effect is mostly *size*: signal grows monotonically with the inventory, coverage of the bigram table collapses beyond about 100–125 signs, and the field's own ∼125-sign catalogue sits where the two criteria cross. Composition matters at fixed size, but differently for different questions: it moves conditional entropy by 0.004 when the frequent core is held fixed and by 0.06–0.14 when the merge rule may touch it, while against the scribes' own substitutions in parallel passages the published catalogue absorbs 40 % of variants where the best mechanical rule of the same size absorbs 18 % and chance 0.6 %. Finally, we place three real syllabaries—Rapa Nui and Māori text syllabified, and Linear B—on the same instrument at matched size and length. At 50–55 signs, rongorongo can be merged so that its normalised conditional entropy is indistinguishable from syllabified Rapa Nui (0.707 vs 0.695), confirming that a small catalogue produces the syllabary's *number*; but its distance from its own shuffle is 2.4–3.6 times smaller than that of every syllabary tested even under the best rongorongo catalogue at each size, and up to ten times smaller under typical ones; the gap is unchanged when run lengths are matched and is flat in sample size. Corrupting 5 % of glyph tokens moves the statistic by 0.011, so transliteration error cannot account for either gap. We conclude that levels of these statistics are properties of the catalogue and the sample; only differences—between readings, and between a corpus and its own null—travel.

*Independent researcher, Helsinki. ipezygj2@gmail.com. Code, results and this draft: https://github.com/ipezygj/rongorongo-catalogue-audit, archived at doi:10.5281/zenodo.21964265.

1 Introduction

Every quantitative claim about an undeciphered script rests on a transliteration, and every transliteration rests on a prior decision about what counts as one sign. For rongorongo, the script of Easter Island, that decision has been made several times with very different results. Barthel [1958] catalogued about 600 glyphs; the CEIPP transliteration inherits that inventory and adds variant letters; Horley [2021] reduces it to about 125 basic glyphs by decomposing ligatures and merging allographs; Pozdniakov and Pozdniakov [2007] argue for a core of 52. These are not competing readings of a few doubtful glyphs. They are different answers to the question of how many symbols the script has, and the answers span a factor of ten.

The statistics used to argue that a sign system is, or is not, linguistic—conditional entropy [Rao et al., 2009, 2010], its critics [Sproat, 2010, 2014], n-gram and Zipfian profiles, hapax ratios—are all computed on the transliterated sign stream. Pozdniakov and Pozdniakov [2007] state the consequence plainly: "it is impossible to make use of statistical data, for clearly the data will differ markedly depending upon the inventory of glyphs chosen." The remark is qualitative, and it appears in a paper whose response to the problem was to build a better catalogue rather than to measure how much the choice moves the numbers. Korovina [2026] runs the same n-gram method on two transliterations of the same inscriptions and obtains different verdicts, but treats the gap as a caveat rather than as the object of study.

This note treats the gap as the object of study. The question is not whether rongorongo is writing. It is: *when a statistic is computed on an undeciphered corpus, how much of the number is the corpus and how much is the catalogue?* We answer it on rongorongo because the catalogue disagreement there is unusually large and the corpus is unusually well digitised, and we answer it in the units in which the debate is now conducted—the length-normalised conditional entropy of Rao et al. [2010]—so that the movement can be read directly against the distances that separate languages from DNA, music and code on their plot.

Our contribution is a set of measurements, made on public data with public code, and a reading of them that we have tried to keep within what the measurements support:

1. The catalogue moves the statistic about four times further than the structure it detects (§4).
2. Most of that movement is catalogue *size*, not composition; there is no preferred size below the published ∼125, and no "knee" at the syllable count (§5).
3. Composition matters, but for a different question: the published catalogue encodes graphic knowledge that no mechanical rule of the same size recovers, visible against the scribes' own substitutions rather than in conditional entropy (§6).
4. A small catalogue makes rongorongo *look* syllabic on the raw axis; on the null-relative axis it has 2.4–3.6 times less sequential structure than real syllabaries at matched size and length, even under its best catalogue (§7).
5. Transliteration error, sample size and estimator bias are all real, and all an order of magnitude smaller than the catalogue effect (§8).

2 Data

Corpus. The CEIPP transliteration as maintained and published in XML by Philip Spaelti[1], one file per object, one `<glyph>` element per coded glyph with a Barthel/CEIPP code (e.g. `022bfy`) and a link marker. 25 objects. Line breaks, lacunae, illegible glyphs and end-of-text

[1] `https://kohaumotu.org/Rongorongo/xml/`

markers cut the sign chain: a bigram is counted only when the object attests the adjacency. Under the numeric reading (variant letters stripped, ligatures split) the corpus has 14 812 tokens over 633 types in 1 182 uninterrupted runs.

We verified against the fixed-column sample CEIPP publishes on its own site that the widely used `rongopy` tablet files are this transliteration reformatted (15/15 glyphs of the published sample match). Provenance therefore reduces to CEIPP; the "readings" below are published *granularities* of one transliteration, not independent editions. That is a weaker comparison than three independent editions would be, and it is the one available.

Readings. Three levels that correspond to published positions on what counts as one sign:

- **full**: CEIPP codes with variant letters, ligatures split. $L = 1\,897$ types.
- **base**: numeric Barthel code only. $L = 633$.
- **horley**: Barthel → Horley basic-glyph map as shipped with `rongopy` (638 keys → 126 basic glyphs, 372 keys decomposing a ligature into parts). Codes the map does not reach, or sends to "?", break the chain like a lacuna. Coverage 91.2 % of coded glyphs. $L = 125$, 15 415 tokens (decomposition increases the count).

Reference syllabaries. (§7.) Rapa Nui: the New Testament (gospels and Acts, 117 chapters, Wycliffe translation), syllabified mechanically as (C)V with C $\in$ {p t k ' m n ŋ v r h}, V $\in$ {a e i o u}, long vowels either kept ($L = 105$) or folded ($L = 55$); 322 752 syllables. Māori: Paipera Tapu New Testament (ebible.org), same procedure, C $\in$ {h k m n ng p r t w wh}, $L = 98$ / 52; 460 155 syllables. Linear B: the `linearb.xyz` corpus, 5 889 inscriptions, syllabograms only, word dividers dropped (rongorongo has none), ideograms and numerals cutting the chain; 47 364 signs, $L = 95$. All three are measured on twenty random contiguous windows of exactly the rongorongo token count so that coverage and null distance are compared at matched N. The Māori text is public domain; the Linear B corpus is taken from its public repository; the Rapa Nui text is copyright Wycliffe Bible Translators and is used for measurement only, fetched by the reader from its publisher and never redistributed.

Two codes that are not signs. CEIPP code 000, written `000!` in 380 of its 390 occurrences, marks a glyph that is present on the object but which the edition declines to identify. Code 999 is the vertical divider of the Santiago Staff: 97 tokens, all on that one object, and identified as a divider rather than a sign in Sproat's string-matching pages. The Horley reading drops both, because the map returns nothing for them and unmapped codes break the chain like a lacuna. The numeric and full-variant readings, as published in versions 1 and 2, kept them, so 000 entered those readings as the sixth commonest "sign" in the corpus. Treating both as breaks in every reading (`pseudoglyph_audit.py`) moves the ladder spread of §4 from 0.318 to 0.327 and the ratio from 4.14 to 4.28. The published figures are therefore conservative, and are retained below so that the two versions can be compared; the corrected pair is given here.

3 Method

Statistic. For a sign stream with unigram counts c_i and bigram counts c_{ij} over L observed types, the plug-in conditional entropy is $H_\mathrm{c} = H(\text{bigram}) - H(\text{unigram})$ in bits. Following Rao et al. [2010], who take the logarithm to base L "to compare sequences over different alphabet sizes," we report $\hat{H}_\mathrm{c} = H_\mathrm{c}/\log_2 L$, which lies in $[0, 1]$ between their Min Ent and Max Ent references. Every distinction their plot draws lives on this axis. Raw bits would make catalogue choice look damaging for free, because changing L changes the ceiling; the normalisation is what divides that out, and the question is whether the movement survives it.

reading	L	tokens	coverage	$\hat{H}_c$	signal	z
full (CEIPP + variants)	1 897	14 812	0.25 %	0.385	−0.046	−70
base (CEIPP numeric)	633	14 812	1.74 %	0.522	−0.055	−68
horley (published ∼125)	125	15 415	23.9 %	0.703	−0.077	−67
spread across readings				0.318		
largest structural signal					0.077	
ratio					4.1×	

Table 1: Three published granularities of the same 25 inscriptions on the Rao–Sproat scale. Miller–Madow correction applied to real and null alike gives spread 0.309, signal 0.083, ratio 3.7×, and does not reorder the readings.

Null. A unigram-matched shuffle: tokens permuted across the corpus, run lengths preserved, so the unigram distribution and L are identical and all sequential structure is destroyed. 200 permutations per reading (100 in the sweep, 30–50 in the reference runs). The *signal* is $\hat{H}_c(\text{real}) - \overline{\hat{H}_c(\text{null})}$; z is that difference over the null standard deviation. Negative signal means the corpus is more predictable than its shuffle.

Coverage. The fraction of the $L \times L$ bigram table with at least one observation. It says whether H_c is estimable at all at that L and N.

Merge rules (sweep). Starting from the 633 numeric codes, a catalogue of size L is produced by one of: *numeric*, sort by Barthel number and cut into L contiguous classes of equal type count (Barthel's numbering follows shape, so this is the cheapest shape-based rule); *horley*, the published 125 merged further downward by the same numeric cut; *freq*, keep the $L-1$ most frequent codes and pool the rest; *random*, L classes of equal type count drawn at random (20 draws), which is what a catalogue of size L with no information in its composition does. Everything above the random line at a given L is what composition adds; the line itself is what size does.

Code. All scripts are standard-library Python and are released at `https://github.com/ipezygj/rongorongo-catalogue-audit` (archived: doi:10.5281/zenodo.21964265), with a fetch script for every input (nothing third-party is redistributed). Random seeds are fixed; every number below is reproducible from the public XML.

4 The catalogue moves the instrument four times further than the signal

Table 1 is the headline. Structure is present under every reading: the corpus is 59–97 standard deviations more predictable than its shuffle. What does not survive the change of reading is any statement about *how much.* Moving from the variant-level reading to Horley's moves $\hat{H}_c$ by 0.318—nearly a third of the whole axis on which Rao et al. [2010] separate languages from non-linguistic systems—while the signal itself is 0.077.

The sharpest single observation is in the coverage column. Under the variant reading 0.25 % of the bigram table is observed and the script looks highly structured (0.385); under Horley's 23.9 % is observed and it looks much closer to Max Ent (0.703). *The more estimable the statistic, the less language-like the script appears.* High "structure" under fine-grained readings is, at least in part, sparse-table bias, not a property of the writing.

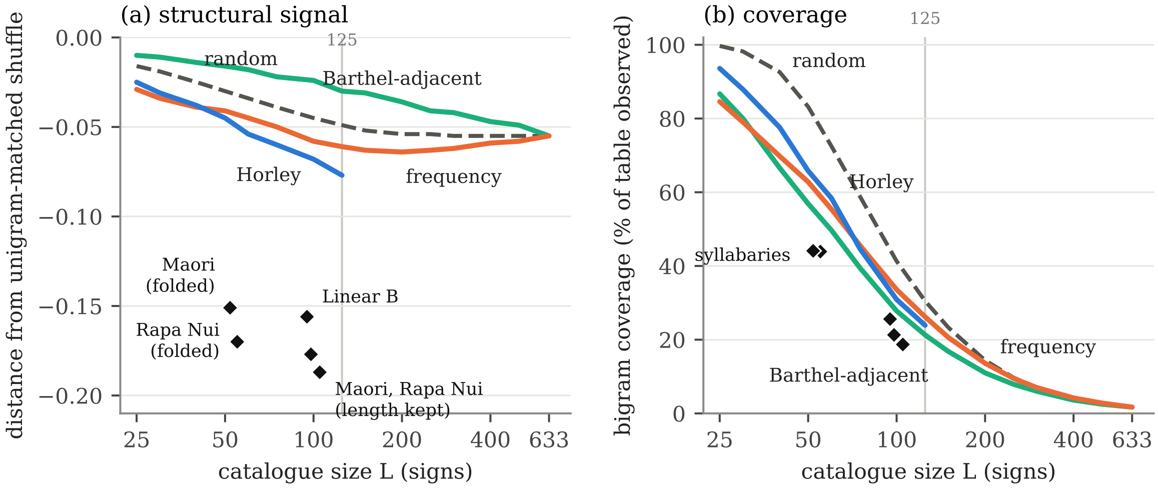

Figure 1: Rongorongo under four merge rules as a function of catalogue size (lines), and real syllabaries at their own size (diamonds; Rapa Nui and Māori syllabified with vowel length folded or kept, and Linear B), all at the rongorongo token count. (a) Distance of the corpus from its unigram-matched shuffle on the log-base-L scale: more negative is more sequential structure. No rule has an interior optimum except Horley's published 125; every syllabary sits 2.4–3.6 times further from its shuffle than the best rongorongo catalogue at the same L. (b) Share of the $L \times L$ bigram table observed. The grey line marks $L = 125$.

Sample size. Rao et al.'s Indus corpus has about 7 000 sign occurrences over 417 types, 16.8 tokens per type; the CEIPP numeric reading has 23.4. Rao et al. note that "beyond $N = 6$ the entropy estimates become less reliable due to the small sample size"; on random sub-samples of the present corpus, H_c is still rising at 100 % of the data, and under the variant reading the last quarter of the corpus contributes 83 % of the whole structural signal. The estimate has not converged at $N = 2$.

The denominator is itself contested. Rao et al. [2010] normalise with $L = 417$ (Mahadevan). Parpola gives 386, ICIT 584, Wells 694. Changing only the denominator, corpus untouched, moves a value near 0.5 by about 0.045—the size of the entire structural signal we measure in rongorongo—and that is a lower bound, since a change of sign list changes the string as well.

5 Size, not composition: a sweep over catalogue length

Korovina (p.c., 2026) named two explanations for the gap that she could not separate: that a longer catalogue simply performs worse because its length exceeds what the corpus can populate, or that two reasonable catalogues of the same size can give substantively different answers—"two very different things from a theoretical standpoint." The sweep is designed to separate them.

Three things follow from Table 2 and Figure 1(a).

There is no knee at the syllable count. Under every rule the distance from the shuffle grows monotonically with L; the only interior maximum anywhere in the grid is Horley's published 125 (-0.077, $z \approx -77$), and merging Horley downward loses signal steadily (-0.045 at 50, -0.025 at 25). Coverage, going the other way, falls below a quarter of the table once L passes 100–125. If "best" means *most structure visible while the statistic is still estimable*, the two criteria cross at roughly 100–125—where the field's own catalogue sits—and not at 50.

	numeric		horley		freq		random	
L	signal	cov	signal	cov	signal	cov	signal	cov
25	−0.010	87 %	−0.025	94 %	−0.029	85 %	−0.016	100 %
50	−0.016	57 %	−0.045	66 %	−0.041	63 %	−0.030	83 %
75	−0.022	40 %	−0.060	45 %	−0.050	46 %	−0.039	59 %
100	−0.024	28 %	−0.068	31 %	−0.058	34 %	−0.045	41 %
125	−0.030	21 %	**−0.077**	24 %	−0.061	26 %	−0.049	31 %
200	−0.036	11 %	—	—	−0.064	14 %	−0.054	14 %
300	−0.042	6 %	—	—	−0.062	7 %	−0.055	7 %
633	−0.055	1.7 %	—	—	−0.055	1.7 %	−0.055	1.7 %

Table 2: Structural signal and bigram coverage as a function of catalogue size, four merge rules, same corpus. Full grid (15 sizes, z, $\hat{H}_c$) in the supplement.

Size dominates. At $L = 633$ every rule is the identity and the four columns coincide; as L shrinks, all four lose signal together, and the spread between rules at any fixed L (0.02–0.03) is smaller than the movement along L (0.05–0.06 between 25 and 633 for the same rule). Korovina's first hypothesis is the larger effect.

But composition is not zero, and its sign is informative. At every $L \leq 125$ the order is horley > freq > random > numeric. Horley beats random partitions of the same size by 5–11 random-draw standard deviations; frequency by 4–7; and the numeric rule—adjacent Barthel numbers merged—is *worse than random* by 3–10. Barthel's numbering follows shape, so this says that shape-adjacent signs behave differently in sequence, and merging them destroys structure that a random merge preserves by accident. §6 pursues this.

6 Composition: what the published catalogue knows

At fixed L, holding the core. Korovina describes the field as agreeing on some 40–50 frequent signs and disagreeing about the rest. Holding the 50 most frequent numeric codes distinct (64.4 % of tokens) and cutting the 583-sign tail into 75 classes by three defensible rules (numeric adjacency, frequency, leading-digit family) at exactly $L = 125$ gives $\hat{H}_c$ of 0.758, 0.754 and 0.756; 300 arbitrary regroupings of the same tail give 0.7632 ± 0.0008. Composition alone, core held, moves the statistic by 0.004—one per cent of the length effect—though all three defensible rules sit below the arbitrary ones by $z = -6.6$ to -10.9, so the merge rule carries information.

At fixed L, rule free to touch the core. When the rule may merge frequent signs (§5), the spread across rules at fixed L is 0.06 at 125 and 0.14 at 50. Both numbers stand; they answer different questions. The first says that if researchers agree on the frequent core, their disagreement about the tail barely registers in conditional entropy. The second says that if they do not, it does.

Against the scribes. Conditional entropy is one question to ask of a catalogue. Another is whether it groups the signs the scribes themselves interchanged. Where the same passage recurs on different objects with a one-sign difference, the two signs at that position are candidates for allographs—the evidence Pozdniakov used by hand in 1996. Seed-and-extend alignment (exact 5-mer seeds between lines on different objects, then alignment, every one-for-one mismatch recorded) yields 119 cross-object passage pairs, 164 distinct substitution pairs, 266 events; the frequent ones—(002,021), (400,600), (045,046), (254,256), (093,095)—are numerically adjacent, which given Barthel's shape ordering is independent evidence that the alignments find variants

		reference ($N = 14\,812$ windows)			rongorongo at same L	
reference	L	$\hat{H}_c$	signal	cov	$\hat{H}_c$ range	signal range
Māori, length folded	52	0.684	−0.151	44 %	0.70–0.88	−0.015 to −0.046
Rapa Nui, length folded	55	0.695	−0.170	44 %	0.71–0.87	−0.017 to −0.047
Linear B	95	0.668	−0.156	26 %	0.71–0.79	−0.026 to −0.063
Māori, length kept	98	0.598	−0.177	21 %	0.70–0.79	−0.024 to −0.065
Rapa Nui, length kept	105	0.598	−0.187	19 %	0.70–0.78	−0.026 to −0.068

Table 3: Real syllabaries versus rongorongo merged to exactly the same L under the four rules of §5. Reference values are means over 20 windows of the rongorongo token count (s.d. of signal 0.007–0.012). Rongorongo ranges span numeric, horley, freq and random; the freq rule gives 0.707 at $L = 55$.

rather than noise. Scoring catalogues on how often they place both members of a substitution in one class, all at $L = 125$, on the same 139 events the Horley map can classify:

Horley (published)	40.3 %
numeric proximity	18.0 %
leading-digit family	10.1 %
frequency	0.7 %
random, 300 draws	0.6 ± 1.1 %

Horley sits 35 standard deviations above chance and 2.2 times the best mechanical rule of the same size (a first pass that scored Horley on 139 events and the mechanical rules on 266 gave 2.9×; the common set is the fair one). Composition, then, is the *whole* effect where the scribes are the witnesses, and one per cent of it where conditional entropy is. The published catalogue encodes graphic judgement that no mechanical rule recovers; it simply does not surface in the statistic the debate has used. Limits: substitution pairs are candidates, not established allographs (two different words can differ by one sign); seeds require an exact match, so only variation in otherwise identical context is visible; 60 % of substitutions are not absorbed by Horley and we cannot tell whether the catalogue is still too fine or those passages differ in wording; and we find 119 cross-object pairs where Horley catalogued 146 passages by hand—the right order, not the same number.

7 Does a small catalogue make rongorongo look syllabic?

Korovina (p.c.) predicted that reducing the catalogue to about 50—the number of Rapa Nui syllables—"will inevitably get more similarity to syllables than to anything else," as a property of the material rather than a bug. Testing this needs syllabaries on the same instrument, at the same L and the same N. Table 3 gives them; Figure 1 places them on the sweep.

The prediction is right on the axis the Rao–Sproat exchange is conducted on. At $L = 55$ the frequency-merged rongorongo comes out at 0.707 and syllabified Rapa Nui at 0.695: indistinguishable. Under the other rules the same corpus lands anywhere from 0.73 to 0.87. So at fifty the number can sit on the syllabary value or well away from it, and which happens depends only on which fifty. That is the "property of the material" Korovina describes, with a figure attached, and it is a second way of saying what §4 said: the level of $\hat{H}_c$ is not a property of the corpus.

The distance from the shuffle separates them. Every real syllabary—Rapa Nui, Māori, Linear B—sits 0.15 to 0.19 below its own shuffle at matched L and N. Rongorongo, under every catalogue from 52 to 105 and every rule, sits 0.02 to 0.07 below its own; and its bigram table is fuller (51–78 % of cells observed against 44 %): the signs combine more freely than syllables in a language do. A fifty-sign rongorongo can wear the syllabary's number, but it has less sequential structure than syllabic text of the same alphabet and length: 2.4–3.6 times less against the best rongorongo catalogue at each size (Linear B 2.4, Rapa Nui with length 2.8, Māori 3.3, Rapa Nui folded 3.6), and up to ten times less against the other rules. Linear B is the figure to put in front of a sceptic: an administrative script of lists and tallies, and it still comes out at -0.152 against rongorongo's best -0.064 at the same L.

Two checks a referee would ask for. *Run length.* The rongorongo chain is cut by lacunae and line ends into 1 182 runs of mean 12.5 tokens; the references were cut at verse and inscription boundaries, and the Polynesian ones are longer (mean 28–36). Re-cutting each reference window into exactly the rongorongo run-length multiset before measuring moves the Polynesian signals by less than 0.01 (Rapa Nui $-0.175 \to -0.167$, Māori $-0.152 \to -0.146$). Linear B runs are *shorter* (mean 3.9), so the legitimate direction is the reverse: cutting rongorongo into Linear B's run lengths moves its signal from -0.064 to -0.062. (Cutting Linear B into rongorongo's longer runs would splice inscriptions together and invent adjacencies; it lowers the Linear B figure to -0.107, and we do not use it.) *Sample size.* Measured at 25, 50, 75 and 100 % of the rongorongo token count with run lengths matched at each N, every series is flat: rongorongo Horley-125 goes -0.066, -0.070, -0.075, -0.077; Horley merged to 55 stays between -0.045 and -0.048; the references stay within their window-to-window scatter. The gap is not a sample-size artefact and would not close with more tablets; rongorongo is already near its asymptote at a quarter of the corpus.

Native prose. The references above are Bible translations, which is what was reachable. Korovina (p.c., 22 August 2026) then supplied four collections of native Rapa Nui text—Campbell's oral-tradition corpus, Englert's texts, the Isla6 edition and Manuscript E—giving 167 077 syllable tokens after Spanish and editorial material is removed at word level, each removal cutting the chain (`korovina_ref.py`). They confirm the gap. At matched L and N, Englert gives -0.157, Isla6 -0.156, Manuscript E -0.154 and the pooled corpus -0.145, against rongorongo's best of -0.045 to -0.066 at the same sizes: ratios of 2.4 to 3.5, inside the range measured on the translations. Native prose gives a slightly *weaker* signal than the Bible does, the translation being the more formulaic text, so the earlier reference was if anything flattering to the language side. The coincidence of level survives too: pooled native prose at $L = 55$ sits at $\hat{H}_c = 0.712$ where frequency-merged rongorongo sits at 0.707.

One collection dissents, and it is the instrument again. Campbell's corpus gives -0.083, a ratio of 1.8 rather than 3.5. It is the only one of the four with neither macrons nor systematic glottal marking—glottal stops appear 3.0 times per thousand characters against Isla6's 46.1—and the glottal is a phoneme in Rapa Nui, so leaving it unwritten merges syllables that the language distinguishes. Degrading Isla6, the best-marked collection, in one respect at a time and remeasuring (`campbell_probe.py`): stripping its glottals and macrons moves its signal from -0.165 to -0.130, which is 43 % of the distance to Campbell; truncating it to Campbell's token count moves it 26 %; cutting its chain at Campbell's word-drop rate moves it 5 %. Three quarters of the dissent is edition and sample size rather than language. We report this not as a caveat but as the argument of §4 arriving where it was not expected: two editions of one living language, on one instrument, differ by a factor of two, and the difference is in the transcription.

We do not draw a conclusion about the script from this. It is consistent with rongorongo not being a plain syllabary of Rapa Nui, and also with a syllabary heavily obscured by ligature, allography and layout conventions our merge rules do not model. What it rules out is that the resemblance at $L \approx 50$ on the raw axis is evidence of syllabic structure: the axis cannot tell the two apart, and the axis that can does not show it. Limits: the references are single-genre

	rongorongo (CEIPP, Horley)	Indus (Mahadevan 1977)
tokens / mean run	14 812 / 12.5	13 372 / 3.7
best structural signal (z)	−0.077 (−77)	−0.217 (−169)
vs. syllabified Rapa Nui, same N, run lengths, L	2.4–3.6× less	0.7× (more)
AA / ABA vs. shuffle	1.6× / 3.8× (seeks)	0.9× / 0.33× (avoids)
$\hat{H}_c$ span, full size sweep	0.318 (125–1 897) = 4.1× signal	0.384 (25–417) = 1.8×
$\hat{H}_c$ span, published catalogues	catalogues 50–600 (10×)	catalogues 386–702 (1.8×); 300–417 measured: 0.16× signal
sweep shape	monotone, no knee	interior optimum $L \approx 150$–200
shape-ordered merge vs random	worse than random (−3 to −10 sd)	equal (−1 sd)

Table 4: The same instrument on two undeciphered corpora. Code and results: `indus/`.

prose against 25 heterogeneous objects, and the Polynesian ones divide into translations and native collections that agree with each other only to within the factor of two discussed above; the (C)V syllabification is mechanical and does not separate diphthongs; Linear B is the only reference that is itself an ancient inscribed corpus, which is why we lean on it.

8 Noise, sample and estimator: real, and an order of magnitude smaller

Transliteration error. Korovina notes that all existing transliterations are "dirty"—contain errors, not merely recodings—"although the error rate is not very high." Replacing a fraction p of glyph tokens by another sign drawn from the corpus unigram distribution (20 draws each) moves $\hat{H}_c$ by +0.002, +0.006, +0.010, +0.020 at $p = 1, 3, 5, 10\,\%$ under the numeric reading and +0.003, +0.008, +0.013, +0.024 under Horley's; the structural signal erodes by the same amounts. Five per cent error is about a thirtieth of the catalogue-length effect and a sixth of the signal; ten per cent removes a third of the signal and leaves the rest. It cannot account for the spread in Table 1, nor for the gap in Table 3. The error model is simple—errors do not preferentially fall on look-alike signs—and is offered as an order of magnitude.

Estimator. The plug-in estimator is biased downward on sparse tables. Miller–Madow correction, applied to unigram and bigram before subtraction and to real and null alike, changes the ratio of §4 from 4.1 to 3.7, does not reorder the readings, and barely shrinks the spread (0.318 → 0.309); the signal grows slightly (0.077 → 0.083) because the null moves more than the data. But Miller–Madow assumes $m/N \to 0$, and here the number of observed bigram types over bigram tokens is 0.67, 0.51 and 0.27 for the three readings—most bigrams are seen once. In that regime no plug-in-family estimator is reliable, and corrected levels should not be quoted as absolute either. *Differences*—between readings, and between data and null—survive because the bias is on both sides.

9 What follows

1. *Levels do not travel; differences do.* A normalised conditional entropy of 0.4 or 0.7 for an undeciphered script is a statement about the catalogue and the sample before it is a statement about the script. What can be compared across corpora is the corpus's distance from its own unigram-matched null, and even that only at matched L and N.

2. *Report the catalogue as a parameter.* Any statistic on an undeciphered corpus should be reported as a curve over catalogue size, or at least at two published granularities, with

bigram coverage alongside. A single number at a single L invites the reader to compare it with a language plotted at another L, and §7 shows how misleading that comparison can be.

3. *The field's catalogue is where the instrument points.* The ∼125-sign inventory reached by hand is where structure is maximal and the table still a quarter populated. That is a small vindication of the epigraphers, and a reason not to expect statistics to improve on their judgement about *size*; what statistics can add is a test of *composition*, and there the scribes' substitutions, not conditional entropy, are the evidence.

4. *The same instrument gives the opposite verdict on Indus—which is the point.* We ran the identical code on the Indus corpus (Mahadevan's 1977 concordance as digitised by Hamilchin [2026]: 3 574 text lines, 13 372 sign tokens, 417 signs, uncertain readings and lost signs breaking the chain; and the 1 548-text EBUDS set of Rao et al. [2009], whose normalised bigram figure we reproduce at 0.387). Table 4 sets it beside rongorongo. Indus texts are short (mean 3.7 signs) and rigidly positional—the terminal "jar" sign 342 is a tenth of all tokens—and the corpus sits 0.217 below its shuffle ($z \approx -169$), *further* than syllabified Rapa Nui, Māori or Linear B at the same N and run lengths; it avoids immediate repetition (AA 0.9, ABA 0.33 of expectation) where rongorongo seeks it. Sweeping catalogue size from 25 to 417 moves $\hat{H}_c$ by 0.384, 1.8 times the signal; but the *published* catalogues disagree only 1.8-fold (Parpola 386, Mahadevan 417, ICIT 702), and across the range we can reach by merging (300–417) the movement is 0.035, a sixth of the signal. Keeping Mahadevan's uncertain readings as their own types (561 vs 417) moves it by 0.02. Mahadevan's numbering, unlike Barthel's, carries no sequential information (merging by it equals random within one standard deviation). So the catalogue problem is script-specific: on Indus the signal is large and the catalogues close, and the instrument is safe; on rongorongo the signal is small and the catalogues span an order of magnitude, and it is not. Nair [2026], working at one ICIT granularity, writes that "no fully validated cross-concordance exists" and that his 584 "should not be directly compared" with 417 or 386; the numbers above say how much that caution is worth in each direction. What Indus's large signal *means* is a separate question—it is positional structure of very short texts, which Sproat [2014] argues non-linguistic systems also show—and we do not enter it here.

Acknowledgements

I am grateful to Evgeniya Korovina (Institute of Linguistics, Russian Academy of Sciences) for a correspondence in August 2026 that shaped this note: for naming the length-versus-composition distinction, for the question about catalogue size that became §5–7, for the observation about transliteration error, for reading a draft, and for the four collections of Rapa Nui prose on which the second half of §efsec:syllabary now rests. Philip Spaelti's XML edition of the CEIPP corpus, the `rongopy` project's Horley map, ebible.org, and the `linearb.xyz` corpus made the measurements possible. Errors of palaeography and of reasoning are mine.

A A calibrated decipherment attempt

Since the natural objection to §7 is that a bare conditional entropy might miss a syllabic mapping that a decipherer would find, we ran the obvious one and calibrated it first. Model: a many-to-one mapping from the 125 Horley glyphs to the 55 Rapa Nui syllables, scored by an add-k bigram model of syllabified Rapa Nui (247k syllables; 20 % held out) plus a channel term $\sum_g n_g \log(n_g/N_{m(g)})$ that charges for homophony (without it the search collapses everything onto two vowels and "beats" the true mapping). Search: Metropolis with annealing, 300k proposals, six restarts; posterior width read as restart agreement per glyph. Everything at the rongorongo token count and run lengths. Held-out Rapa Nui enciphered by a random 1:1 substitution is recovered exactly (100 % of tokens, two of six restarts reach the true-mapping likelihood); a homophonic cipher with 99 symbols is recovered to 37 % of tokens, with a likelihood gain of +0.35 nats per token over its own shuffle. Rongorongo under its best mapping gains +0.17 over its shuffle ($z \approx 21$)—and Linear B, which is Greek, pushed through the same pipe gains +0.15; Māori, a sister language, +0.27. The decipherer reads rongorongo as Rapa Nui exactly as well as it reads Linear B as Rapa Nui: it is finding sequential structure, not language. Only one glyph locks across restarts (Horley 76 → 'a, 6/6); the rest sit at 1–4 of 6. This does not exclude a logographic-plus-syllabic mixture, ligatures as words, or an anchored attack via the Mamari calendar; it does say that a plain one-glyph-one-syllable hypothesis under a bigram model has no more support from the corpus than an unrelated script would give it. Code: `decipher.py`, results in `results/decipher_results.txt`. We also tried the one anchor the corpus offers, the lunar calendar of Mamari (Ca6–9; Guy, 1990): the text has exactly 28 crescents (Barthel 040) in groups of 2/6/3/7/3/5/2, a heralding sequence repeated eight times, and the full-moon glyph 152 near the middle, as Guy describes. Clamping glyph 040 to each of the 55 syllables in turn and re-running the global search ranks the candidates as noise—*ma* (mahina, marama) 6th, *hi* 19th, *po* (night) 30th of 55, total spread 0.11 nats per token—and the herald reads under the best mapping as a string the language model likes no better than chance ($z = 1.7$). The calendar supports the crescent as a pictogram, not a syllable. It also supplies one more instance of the paper's point: the Horley map folds 040 and 041 into one class and does not reach 152 at all, so the distinction Guy reads—night crescents against herald crescents, and the full moon—is invisible under the very catalogue on which the statistics are computed. Code: `mamari.py`. A mixed model (each glyph a syllable or one of 203 frequent Rapa Nui words, syllable-LM score normalised to the mean per-syllable log-probability, with a logogram prior) was also built and *failed its own control*: on held-out Rapa Nui with thirty frequent words replaced by logogram symbols it recovers at most 4.5 % of tokens at any prior, so its verdict on rongorongo (same pattern: gain over shuffle 0.10 against Linear B's 0.11 and the control's 0.14; moon and night words for the crescent all score below the free choice) carries no evidential weight either way. We report it as a limit of the instrument at this corpus size, not as a finding about the script (`mixed.py`).

A repetition profile, and the null it turned out to depend on. Under any one-to-one key the rates of immediate repetition (AA), of triple repetition (AAA) and of period-two repetition ($ABAB$) in the glyph stream equal the rates in the syllable stream, so a repetition profile is inherited by every syllabic reading whatever the key. That much still holds, and it is why the measurement seemed worth making. Versions 1 and 2 of this paper reported rongorongo at AA 1.6–2.4, AAA 5–15 and $ABAB$ 30–100 times expectation, against syllabified Rapa Nui and Māori at AA 0.5–0.6, and concluded that a plain syllabic reading of a Polynesian language was ruled out whatever the key. Those figures were computed against a shuffle of the *pooled* corpus. The corpus is 25 objects with different vocabularies, so its pooled unigram distribution is a mixture, and any locally concentrated stream looks repetition-seeking against a mixture. The reference languages were each a single homogeneous text, where no such gap between pooled and local exists. The comparison was therefore not like for like, and the conclusion does not survive the correction.

The references do not care which null is used—syllabified Rapa Nui sits at 0.57 on AA under all three—because each is one homogeneous text. Rongorongo cares a great deal. Under the matched null it sits at AA 1.12 against the languages' 0.52–0.74, at AAA 1.68, and at $ABAB$ 20.7 against 4.7–9.3. Two things follow. *The elimination fails*: Linear B is an actual syllabary and sits at AAA 4.33, higher than rongorongo, so a syllabary can produce a profile

	shuffle pooled corpus			shuffle within each text			shuffle within each run		
	AA	*AAA*	*ABAB*	*AA*	*AAA*	*ABAB*	*AA*	*AAA*	*ABAB*
rongorongo	1.59	4.34	34.6	**1.12**	**1.68**	**20.7**	0.62	0.63	4.7
Rapa Nui (NT)	0.57	0.47	8.3	0.57	0.41	9.3	0.57	0.55	8.1
Rapa Nui (Isla6)	0.69	0.29	7.7	0.69	0.31	8.0	0.65	0.25	5.8
Māori (NT)	0.52	0.19	4.8	0.52	0.16	4.7	0.52	0.20	3.5
Linear B	0.72	4.38	1.2	0.74	4.33	1.1	0.64	2.56	0.8

Table 5: Observed repetition divided by its expectation under three nulls (`null_choice3.py`). References are re-cut to the rongorongo run-length multiset before measuring. The middle block is the null that matches how the languages were measured: one shuffle per text, one object being one text.

of this kind and the argument that ruled one out is withdrawn. *What is real is smaller*: on *ABAB* rongorongo stands above every reference under every null, but by a factor of about two rather than of thirty.

The movement is worth more than what it was meant to prove. The languages are null-independent and rongorongo is not, and that is itself a measurement: rongorongo's repetition statistics are governed by local vocabulary concentration rather than by adjacency. Within a run of about twelve signs, shuffling that run's own tokens produces *more* immediate repetition than is observed—813 expected against 496—while doing the same to twelve syllables of Rapa Nui changes nothing at all. The signs of a rongorongo passage are drawn from a much narrower local pool than the syllables of a Rapa Nui passage are, and the adjacencies within that pool are, if anything, avoided. That is stable under every null tried here, it is consistent with the list, tally and litany readings the epigraphers favour, and it is not evidence against a syllabary.

The Santiago Staff, measured rather than read. Fischer [1995] read the Staff as procreation triads X–76–Y–Z, noting himself that glyph 76 "is attached to the first glyph in each section". The reading has been criticised at length since—by Guy, by Pozdniakov and Pozdniakov [2007], and by Robinson, who counted only 63 of 113 sequences obeying the structure—and Sproat [n.d.] removed 76 from the corpus and recomputed approximate string matches across objects, finding that the Staff remained an isolate. We have read the last of these directly and take the others from its summary and from Pozdniakov and Pozdniakov [2007]. The measurement here is a different quantity, repetition within each object against that object's own shuffle (`fingerprint2.py`). Every other object avoids immediate repetition at 0.43–0.83 of expectation. The Staff sits at 0.47 and, uniquely, is depressed at lag 2 as well (0.80). Removing glyph 76—24.3 % of the Staff's tokens after the divider 999 is excluded, against 0.6 % of the rest of the corpus—returns it to the corpus range at lag 1 (0.85) and lag 2 (1.25). The Staff's statistical isolation on this axis is an artefact of one very frequent glyph, and Fischer's triads leave no trace at the period they predict: lag 3 is 0.94 as encoded and 1.09 with 76 removed.

References

T. S. Barthel. *Grundlagen zur Entzifferung der Osterinselschrift.* Cram, de Gruyter, Hamburg, 1958.

S. R. Fischer. Preliminary evidence for cosmogonic texts in Rapanui's rongorongo inscriptions. *Journal of the Polynesian Society*, 104(3):303–321, 1995.

R. Sproat. Approximate string matches in the rongorongo corpus. `https://richardsproat.com/ror/`, accessed August 2026.

J. B. M. Guy. On the lunar calendar of tablet Mamari. *Journal de la Société des Océanistes*, 91:135–149, 1990.

Hamilchin. `indus-cipherable`: digitisation of Mahadevan (1977) and the EBUDS set. `https://github.com/Hamilchin/indus-cipherable`, accessed August 2026.

P. Horley. Allographic variations and statistical analysis of the Rongorongo script. *Rapa Nui Journal*, 19(2):107–116, 2005.

P. Horley. *Rongorongo: Inscribed Objects from Rapa Nui*. Rapanui Press, Viña del Mar, 2021. (Basic-glyph catalogue of ∼130 signs; the Barthel→Horley map used here is the one distributed with `rongopy`, `https://github.com/jgregoriods/rongopy`.)

E. Korovina. A computational evaluation of syllabic hypotheses for Rongorongo: evidence from n-gram analysis. In *Proceedings of the Fourth Workshop on Language Technologies for Historical and Ancient Languages (LT4HALA 2026) @ LREC 2026*, pages 177–183, Palma, 2026. `https://aclanthology.org/2026.lt4hala-1.16/`

G. A. Miller. Note on the bias of information estimates. In *Information Theory in Psychology*, pages 95–100. Free Press, 1955.

I. Mahadevan. *The Indus Script: Texts, Concordance and Tables*. Archaeological Survey of India, New Delhi, 1977.

A. Nair. How non-linguistic is the Indus sign system? A synthetic-baseline scorecard. arXiv:2604.17828, April 2026.

K. Pozdniakov and I. Pozdniakov. Rapanui writing and the Rapanui language: preliminary results of a statistical analysis. *Forum for Anthropology and Culture*, 3:3–36, 2007.

R. P. N. Rao, N. Yadav, M. N. Vahia, H. Joglekar, R. Adhikari, and I. Mahadevan. Entropic evidence for linguistic structure in the Indus script. *Science*, 324(5931):1165, 2009.

R. P. N. Rao, N. Yadav, M. N. Vahia, H. Joglekar, R. Adhikari, and I. Mahadevan. Entropy, the Indus script, and language: a reply to R. Sproat. *Computational Linguistics*, 36(4):795–805, 2010.

R. Sproat. Ancient symbols, computational linguistics, and the reviewing practices of the general science journals. *Computational Linguistics*, 36(3):585–594, 2010.

R. Sproat. A statistical comparison of written language and nonlinguistic symbol systems. *Language*, 90(2):457–481, 2014.